BIO101 MCQS 3

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Southern blotting:

1. **What is the purpose of using restriction endonucleases in Southern blotting?**
   - A) To stain DNA fragments
   - B) To amplify DNA fragments
   - C) To cut DNA into smaller fragments
   - D) To transfer DNA onto a membrane

   Answer: C) To cut DNA into smaller fragments

2. **Which buffer is commonly used for electrophoresis in Southern blotting when running gels for longer durations?**
   - A) TAE
   - B) TBE
   - C) EDTA
   - D) SDS

   Answer: B) TBE

3. **What is the purpose of using ethidium bromide or SYBR green in Southern blotting?**
   - A) To label probes
   - B) To block nonspecific binding
   - C) To stain DNA for visualization
   - D) To amplify DNA fragments

   Answer: C) To stain DNA for visualization

4. **Which of the following is used as a probe in Southern blotting for specific detection?**
   - A) Ethidium bromide
   - B) TAE buffer
   - C) Salmon sperm DNA
   - D) Radioactively labeled RNA

   Answer: D) Radioactively labeled RNA

5. **What is the purpose of the hybridization step in Southern blotting?**
   - A) To cut DNA fragments
   - B) To separate DNA fragments by size
   - C) To bind probes to specific DNA sequences

- D) To visualize DNA fragments

Answer: C) To bind probes to specific DNA sequences

6. **Which of the following is a common step in DNA purification using a commercial kit like GenElute™?**

   - A) Proteinase digestion
   - B) Ethanol precipitation
   - C) PCR amplification
   - D) Electrophoresis

   Answer: A) Proteinase digestion

7. **What is the purpose of using a detergent during cell lysis in DNA extraction?**

   - A) To precipitate DNA
   - B) To denature proteins
   - C) To inhibit RNA degradation
   - D) To remove lipids

   Answer: B) To denature proteins

8. **Which of the following is NOT typically included in a DNA extraction buffer?**

   - A) Tris-HCl
   - B) EDTA
   - C) SDS
   - D) Ethanol

   Answer: D) Ethanol

9. **Which commercial product is commonly used for the isolation of DNA from various sources such as mammalian cells, plants, bacteria, and fungi?**

   - A) GenElute™ DNA Purification Kit
   - B) Taq DNA Polymerase
   - C) Agarose Gel
   - D) Ethidium Bromide

   Answer: A) GenElute™ DNA Purification Kit

10. **What is the advantage of using commercial DNA purification kits over manual methods?**

    - A) Lower cost

- B) Higher purity of DNA
- C) Faster processing time
- D) Less equipment required

Answer: B) Higher purity of DNA

11. Which step is critical for removing proteins from a DNA sample during purification?

- A) Alcohol precipitation
- B) Centrifugation
- C) Proteinase digestion
- D) PCR amplification

Answer: C) Proteinase digestion

12. In DNA purification, what is the purpose of adding ethanol to a DNA sample?

- A) To wash the DNA
- B) To denature proteins
- C) To precipitate DNA
- D) To lyse cells

Answer: C) To precipitate DNA

13. Which of the following buffers is commonly used for DNA electrophoresis?

- A) TAE (Tris-Acetate-EDTA)
- B) SDS (Sodium Dodecyl Sulfate)
- C) SSC (Sodium Chloride-Sodium Citrate)
- D) Tris-HCl

Answer: A) TAE (Tris-Acetate-EDTA)

14. Which component is used to dissolve DNA in the elution step of a DNA purification kit?

- A) Isopropanol
- B) TE buffer
- C) Ethanol
- D) SDS

Answer: B) TE buffer

15. Which enzyme is commonly used to degrade proteins in a DNA extraction process?

- A) DNA polymerase
- B) Proteinase K
- C) RNA polymerase
- D) Restriction endonuclease

Answer: B) Proteinase K

16. **Why is it important to keep all reagents necessary for restriction digestion on ice before setting up the reaction?**

    - A) To prevent enzyme denaturation
    - B) To facilitate DNA denaturation
    - C) To stabilize DNA structure
    - D) To enhance enzyme activity

    Answer: A) To prevent enzyme denaturation

17. **When should the restriction enzyme be added to the digestion reaction mixture?**

    - A) At the beginning of the setup
    - B) Immediately before incubation
    - C) After incubation at 37 °C
    - D) After adding glycerol

    Answer: B) Immediately before incubation

18. **What precaution should be taken to ensure complete digestion of DNA during the restriction process?**

    - A) Add enzyme slowly to avoid bubbles
    - B) Add a surplus amount of enzyme
    - C) Keep samples at room temperature
    - D) Use a lower enzyme concentration

    Answer: B) Add a surplus amount of enzyme

19. **Why is it recommended not to exceed a certain concentration of the enzyme in the digestion reaction?**

    - A) To prevent DNA degradation
    - B) To avoid excess glycerol inhibition
    - C) To reduce pipetting errors
    - D) To speed up the digestion process

    Answer: B) To avoid excess glycerol inhibition

20. Which incubation condition is generally preferred for DNA samples obtained from cloning during restriction digestion?

- A) Overnight at 37 °C
- B) 1–2 hours at 37 °C
- C) 4 hours at room temperature
- D) 30 minutes at 50 °C

Answer: B) 1–2 hours at 37 °C

21. What is the purpose of adding half of the enzyme at the beginning of an overnight digestion reaction and the other half in the morning?

- A) To increase enzyme activity
- B) To reduce enzyme degradation
- C) To ensure complete digestion
- D) To prevent DNA degradation

Answer: C) To ensure complete digestion

22. How can DNA samples be concentrated after digestion for gel electrophoresis?

- A) Adding more restriction enzyme
- B) Precipitation with ethanol
- C) Heating at high temperature
- D) Adding more loading buffer

Answer: B) Precipitation with ethanol

23. What is the purpose of resuspending DNA samples in ddH2O after ethanol precipitation?

- A) To increase DNA concentration
- B) To add color to the samples
- C) To stabilize DNA structure
- D) To remove traces of ethanol

Answer: D) To remove traces of ethanol

24. Why is it important to ensure complete removal of ethanol from DNA samples before loading onto a gel?

- A) To prevent DNA degradation
- B) To prevent well overflow
- C) To increase enzyme activity
- D) To improve gel resolution

Answer: B) To prevent well overflow

25. Which buffer is commonly used for preparing the DNA samples for loading onto a gel after restriction digestion?

- A) SDS buffer
- B) TE buffer
- C) SSC buffer
- D) TBE buffer

Answer: B) TE buffer

26. Why do nucleic acids move towards the anode in an electrophoresis chamber?

- A) Due to their positive charge
- B) Due to electrostatic attraction
- C) Due to their negative charge
- D) Due to magnetic field

Answer: C) Due to their negative charge

27. How is the percentage of the gel determined for gel electrophoresis?

- A) By the size of DNA fragments
- B) By the concentration of agarose
- C) By the concentration of ethidium bromide
- D) By the volume of the buffer

Answer: A) By the size of DNA fragments

28. Why are longer gels preferred for separating genomic DNA or multiple fragments with similar sizes?

- A) To speed up the separation process
- B) To ensure appropriate resolution
- C) To reduce the amount of DNA needed
- D) To decrease the buffer volume

Answer: B) To ensure appropriate resolution

29. When preparing an agarose gel, why should the base gel be poured before placing the comb?

- A) To avoid gel cracking
- B) To prevent DNA degradation
- C) To facilitate even distribution of DNA

- D) To ensure well formation

Answer: D) To ensure well formation

30. How is agarose melted when preparing a gel using a microwave?

- A) Stirring vigorously
- B) Heating in short bursts
- C) Adding salt
- D) Adding ethanol

Answer: B) Heating in short bursts

31. At what temperature is ethidium bromide added to the molten agarose gel?

- A) 4°C
- B) 37°C
- C) 60°C
- D) 100°C

Answer: C) 60°C

32. Why should bubbles be avoided when pouring the agarose gel?

- A) They affect DNA migration
- B) They increase gel transparency
- C) They enhance buffer conductivity
- D) They help in gel setting

Answer: A) They affect DNA migration

33. How is the completion of gel setting usually determined?

- A) By color change
- B) By opalescent appearance
- C) By gel weight
- D) By odor

Answer: B) By opalescent appearance

34. What is the purpose of adding 1X loading buffer to DNA samples before loading them onto the gel?

- A) To increase DNA concentration
- B) To stabilize DNA structure
- C) To add color

- D) To facilitate sample loading

Answer: D) To facilitate sample loading

35. Why are one or two wells left blank when loading DNA samples onto a gel?

- A) To save buffer
- B) To reduce gel thickness
- C) To create reference points
- D) To avoid DNA contamination

Answer: C) To create reference points

36. Why is it necessary to denature DNA during the Southern blotting procedure?

- A) To increase DNA size
- B) To convert DNA to RNA
- C) To produce single-stranded DNA
- D) To decrease DNA concentration

Answer: C) To produce single-stranded DNA

37. What is the purpose of denaturing DNA in an alkaline solution?

- A) To neutralize the DNA
- B) To increase DNA stability
- C) To separate DNA strands
- D) To promote DNA synthesis

Answer: C) To separate DNA strands

38. What type of DNA is formed as a result of the denaturation process?

- A) Double-stranded DNA
- B) Single-stranded DNA
- C) Circular DNA
- D) RNA-DNA hybrid

Answer: B) Single-stranded DNA

39. Denaturation is required for which subsequent step in the Southern blotting process?

- A) DNA replication
- B) Probe hybridization
- C) Gel electrophoresis

- D) Ethidium bromide staining

Answer: B) Probe hybridization

40. In which type of solution is DNA typically denatured for Southern blotting?

- A) Neutral solution
- B) Acidic solution
- C) Alkaline solution
- D) Saline solution

Answer: C) Alkaline solution

41. Which of the following best describes the structure of DNA after the denaturation step?

- A) Helical and double-stranded
- B) Linear and double-stranded
- C) Helical and single-stranded
- D) Linear and single-stranded

Answer: D) Linear and single-stranded

42. What is the main objective of denaturing DNA before probe hybridization in Southern blotting?

- A) To increase probe specificity
- B) To enhance DNA mobility
- C) To facilitate probe binding
- D) To reduce DNA size

Answer: C) To facilitate probe binding

43. During denaturation, which bonds in the DNA molecule are broken?

- A) Phosphodiester bonds
- B) Hydrogen bonds
- C) Covalent bonds
- D) Peptide bonds

Answer: B) Hydrogen bonds

44. What is the effect of an alkaline solution on the DNA double helix?

- A) It enhances the helical structure
- B) It converts DNA to RNA

- C) It separates the two strands
- D) It forms a triple helix structure

Answer: C) It separates the two strands

45. Which step directly follows DNA denaturation in the Southern blotting process?

- A) Gel electrophoresis
- B) Probe hybridization
- C) DNA extraction
- D) DNA staining

Answer: B) Probe hybridization

46. What is the purpose of blotting in the Southern blotting procedure?

- A) To fragment the DNA
- B) To transfer DNA to a membrane
- C) To denature the DNA
- D) To hybridize the DNA

Answer: B) To transfer DNA to a membrane

47. Which type of membranes are commonly used for blotting in Southern blotting?

- A) Paper membranes
- B) Nylon membranes
- C) Plastic membranes
- D) Glass membranes

Answer: B) Nylon membranes

48. What method can be used to permanently fix DNA to the membrane?

- A) Freezing
- B) Heating at 100°C
- C) UV radiation or drying
- D) Centrifugation

Answer: C) UV radiation or drying

49. Which membrane has a higher DNA binding capacity and is more efficient for Southern blotting?

- A) Paper membrane
- B) Nitrocellulose membrane

- C) Nylon membrane
- D) Plastic membrane

Answer: C) Nylon membrane

50. What is the role of NaCl solution in the blotting process?

- A) To denature the DNA
- B) To stain the DNA
- C) To saturate the DNA
- D) To fragment the DNA

Answer: C) To saturate the DNA

51. Which process is described by transferring fragmented DNA to a membrane using electroblotting or capillary blotting?

- A) Hybridization
- B) Electrophoresis
- C) Blotting
- D) PCR

Answer: C) Blotting

52. What is the first step after transferring the gel to a documentation system for capturing a photograph?

- A) Incubating with proteinase
- B) Immersing in electrophoresis buffer mixed with ethidium bromide
- C) Denaturing with alkaline solution
- D) Neutralizing with Tris HCl

Answer: B) Immersing in electrophoresis buffer mixed with ethidium bromide

53. How is DNA transfer efficiency confirmed after blotting?

- A) By staining with Coomassie Blue
- B) By visualizing under UV light
- C) By measuring absorbance at 260 nm
- D) By running an additional electrophoresis

Answer: B) By visualizing under UV light

54. How is DNA fixed on the membrane after blotting in Southern blotting?

- A) By baking at 100°C

- B) By freezing at -20°C
- C) By baking at 80°C or UV crosslinking
- D) By immersing in liquid nitrogen

Answer: C) By baking at 80°C or UV crosslinking

55. What is the final step in the blotting process before proceeding to hybridization?

- A) Immersing the membrane in an acid bath
- B) Baking or UV crosslinking the membrane
- C) Immersing the membrane in ethidium bromide
- D) Running the gel for additional separation

Answer: B) Baking or UV crosslinking the membrane

56. What is the purpose of adding a labeled probe to the membrane buffer during hybridization?

- A) To break down the DNA
- B) To visualize the DNA fragments
- C) To bind to the target DNA sequences
- D) To denature the DNA

Answer: C) To bind to the target DNA sequences

57. What factors influence the time required for the hybridization step?

- A) Temperature and membrane type
- B) Complexity and concentration of the probe
- C) Electrophoresis buffer and gel type
- D) DNA source and ethidium bromide concentration

Answer: B) Complexity and concentration of the probe

58. Which type of probe can be used in the hybridization process?

- A) Only RNA probes
- B) Only DNA probes
- C) Both RNA and DNA probes
- D) Protein probes

Answer: C) Both RNA and DNA probes

59. What is the role of paraffin oil in the hybridization by immersion method?

- A) To act as a solvent for the hybridization solution

- B) To prevent evaporation and maintain a uniform temperature
- C) To denature the DNA
- D) To enhance the probe binding

Answer: B) To prevent evaporation and maintain a uniform temperature

60. What is the primary disadvantage of hybridization by immersion in paraffin oil?

- A) High cost
- B) Uneven background
- C) Low probe specificity
- D) Short hybridization time

Answer: B) Uneven background

61. Why is it important to remove excess probe after hybridization?

- A) To prevent the DNA from denaturing
- B) To reduce non-specific binding
- C) To enhance the mobility of DNA fragments
- D) To increase the concentration of the probe

Answer: B) To reduce non-specific binding

62. What can be done if high background radiation is detected on the membrane after hybridization?

- A) Increase the probe concentration
- B) Perform RNase treatment
- C) Increase the hybridization temperature
- D) Reduce the hybridization time

Answer: B) Perform RNase treatment

63. Which labeled RNA detection method involves wrapping the membrane in plastic and laying it on X-ray film?

- A) Fluorography
- B) 32P-labeled RNA detection
- C) 14C-labeled RNA detection
- D) Autoradiography

Answer: B) 32P-labeled RNA detection

64. How can the specificity of the hybridization be increased during the washing step?

- A) By lowering the temperature
- B) By increasing the salt concentration
- C) By using more detergent
- D) By increasing the number of stringent washes

Answer: D) By increasing the number of stringent washes

65. What determines the duration of the detection step in the hybridization process?

- A) Gel electrophoresis time
- B) Hybridization yield and probe specificity
- C) Type of membrane used
- D) Temperature of the hybridization chamber

Answer: B) Hybridization yield and probe specificity

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1. **What is the primary purpose of Northern blotting?**
    - A) To detect specific DNA sequences in DNA samples
    - B) To detect specific RNA sequences in RNA samples
    - C) To detect specific protein sequences in protein samples
    - D) To detect specific carbohydrate sequences in carbohydrate samples

    Answer: B) To detect specific RNA sequences in RNA samples

2. **What is the first step in the Northern blotting procedure?**
    - A) Hybridization of RNA to probes
    - B) Transfer of RNA to a solid support
    - C) Isolation of intact mRNA
    - D) Detection of the probe molecules

    Answer: C) Isolation of intact mRNA

3. **What type of gel is used for the separation of RNA in Northern blotting?**
    - A) Denaturing agarose gel
    - B) Non-denaturing polyacrylamide gel
    - C) Denaturing polyacrylamide gel
    - D) Non-denaturing agarose gel

    Answer: A) Denaturing agarose gel

4. **What is the purpose of transferring RNA to a solid support in Northern blotting?**
    - A) To amplify the RNA sequences
    - B) To sequence the RNA

- C) To immobilize the RNA for hybridization
- D) To denature the RNA

Answer: C) To immobilize the RNA for hybridization

5. **What is the role of the probes used in Northern blotting?**
   - A) To denature the RNA
   - B) To amplify the RNA
   - C) To bind to complementary RNA sequences
   - D) To visualize the RNA fragments

   Answer: C) To bind to complementary RNA sequences

6. **What step follows hybridization of the immobilized RNA to probes in Northern blotting?**
   - A) Transfer of RNA to a solid support
   - B) Detection of the bound probe molecules
   - C) Removal of non-specifically bound probe molecules
   - D) Isolation of intact mRNA

   Answer: C) Removal of non-specifically bound probe molecules

7. **How are the non-specifically bound probe molecules removed in Northern blotting?**
   - A) By increasing the hybridization temperature
   - B) By washing with a suitable buffer
   - C) By using restriction enzymes
   - D) By decreasing the hybridization temperature

   Answer: B) By washing with a suitable buffer

8. **What is the final step in the Northern blotting procedure?**
   - A) Transfer of RNA to a solid support
   - B) Isolation of intact mRNA
   - C) Detection, capture, and analysis of the bound probe molecules
   - D) Separation of RNA according to size

   Answer: C) Detection, capture, and analysis of the bound probe molecules

9. **Which of the following is NOT a part of the Northern blotting procedure?**
   - A) Isolation of intact mRNA
   - B) Separation of RNA by denaturing gel electrophoresis
   - C) Amplification of RNA sequences
   - D) Hybridization of RNA to complementary probes

   Answer: C) Amplification of RNA sequences

10. **What can be inferred from a successful Northern blot experiment?**
    - A) The presence of a specific protein in the sample
    - B) The presence of a specific DNA sequence in the sample
    - C) The presence of a specific RNA sequence in the sample
    - D) The overall RNA concentration in the sample

    Answer: C) The presence of a specific RNA sequence in the sample

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1. **Which of the following is an application of Northern blotting?**
    - A) Study of gene expression in prokaryotic cells
    - B) Study of gene expression in eukaryotic cells
    - C) Detection of specific DNA sequences
    - D) Detection of specific proteins

    Answer: B) Study of gene expression in eukaryotic cells

2. **Northern blotting can be used to measure the amount and size of which type of molecules?**
    - A) DNA molecules
    - B) Protein molecules
    - C) RNA molecules
    - D) Carbohydrate molecules

    Answer: C) RNA molecules

3. **What is one of the purposes of using a housekeeping gene in Northern blotting?**
    - A) To increase the hybridization temperature
    - B) To amplify RNA sequences
    - C) To normalize samples according to their mRNA content
    - D) To separate RNA fragments by size

    Answer: C) To normalize samples according to their mRNA content

4. **Which of the following is NOT an application of Northern blotting?**
    - A) Estimating the abundance of RNA
    - B) Equalizing the amounts of RNA loaded into lanes of gels
    - C) Measuring the amount of proteins in a sample
    - D) Studying the expression of endogenous constitutively-expressed genes

    Answer: C) Measuring the amount of proteins in a sample

5. **Why is it important to normalize samples in Northern blotting?**
    - A) To ensure that RNA samples do not degrade

- B) To ensure equal loading of RNA into gel lanes
- C) To increase the amount of RNA in each sample
- D) To detect specific DNA sequences

Answer: B) To ensure equal loading of RNA into gel lanes

6. **What does Northern blotting allow researchers to estimate regarding RNA?**
   - A) The sequence of RNA
   - B) The abundance of RNA
   - C) The primary structure of RNA
   - D) The location of RNA within a cell

   Answer: B) The abundance of RNA

7. **Which type of gene is typically used as a reference or control in Northern blotting experiments?**
   - A) Mutated gene
   - B) Exogenous gene
   - C) Housekeeping gene
   - D) Recombinant gene

   Answer: C) Housekeeping gene

8. **What is one of the advantages of using Northern blotting in studying RNAs?**
   - A) It allows for the amplification of RNA sequences
   - B) It provides a quantitative measure of RNA abundance
   - C) It can directly sequence RNA
   - D) It is used for studying protein-protein interactions

   Answer: B) It provides a quantitative measure of RNA abundance

9. **How can Northern blotting help in understanding gene expression?**
   - A) By detecting specific proteins produced by genes
   - B) By measuring the amount and size of RNAs transcribed from genes
   - C) By sequencing the DNA of the gene
   - D) By amplifying the RNA transcribed from the gene

   Answer: B) By measuring the amount and size of RNAs transcribed from genes

10. **Northern blotting is commonly used in molecular biology for which of the following purposes?**
    - A) To detect mutations in DNA sequences
    - B) To measure the levels of specific RNA molecules
    - C) To identify protein-DNA interactions
    - D) To sequence RNA molecules

Answer: B) To measure the levels of specific RNA molecules

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Western blotting:

1. **Western blotting is a technique used to detect the presence of which type of molecules?**
   - A) DNA
   - B) RNA
   - C) Proteins
   - D) Carbohydrates

   Answer: C) Proteins

2. **One of the primary uses of Western blotting is to:**
   - A) Determine the sequence of nucleotides in DNA
   - B) Detect the presence of specific proteins in complex mixtures
   - C) Measure the concentration of glucose in blood
   - D) Sequence RNA molecules

   Answer: B) Detect the presence of specific proteins in complex mixtures

3. **Western blotting can be used to determine:**
   - A) The molecular weight of a protein
   - B) The sequence of a DNA fragment
   - C) The abundance of RNA
   - D) The structure of a carbohydrate

   Answer: A) The molecular weight of a protein

4. **Which of the following is an application of Western blotting?**
   - A) Sequencing DNA
   - B) Measuring relative amounts of a specific protein in complex mixtures
   - C) Amplifying RNA sequences
   - D) Detecting specific mRNA molecules

   Answer: B) Measuring relative amounts of a specific protein in complex mixtures

5. **Western blots are commonly used for the detection of:**
   - A) Viral RNA
   - B) Viral DNA
   - C) Viral proteins
   - D) Viral lipids

Answer: C) Viral proteins

6. **In Western blotting, which of the following is typically used to characterize monoclonal and polyclonal antibody preparations?**
    - A) PCR
    - B) Northern blotting
    - C) Western blotting
    - D) Southern blotting

    Answer: C) Western blotting

7. **Western blotting can help in determining the specificity of the immune response to:**
    - A) Bacterial DNA
    - B) Viral antigens
    - C) Carbohydrate molecules
    - D) Lipid molecules

    Answer: B) Viral antigens

8. **Which step in Western blotting is critical for quantifying the relative amounts of protein present in complex mixtures?**
    - A) Electrophoresis of RNA
    - B) Hybridization with DNA probes
    - C) Detection with labeled antibodies
    - D) Amplification of protein sequences

    Answer: C) Detection with labeled antibodies

9. **What is the significance of determining the molecular weight of a protein using Western blotting?**
    - A) It helps in sequencing the protein
    - B) It identifies the protein based on size
    - C) It measures the RNA content of the sample
    - D) It quantifies DNA concentration

    Answer: B) It identifies the protein based on size

10. **Which of the following is NOT an application of Western blotting?**
    - A) Detecting specific proteins in a sample
    - B) Measuring the abundance of specific proteins
    - C) Determining the molecular weight of proteins
    - D) Sequencing RNA molecules

    Answer: D) Sequencing RNA molecules

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1. **What is the primary purpose of sample preparation in Western blotting?**
    - A. To degrade the proteins
    - B. To extract and prepare proteins for analysis
    - C. To transfer proteins to the blotting membrane
    - D. To block non-specific binding of antibodies

    Answer: B. To extract and prepare proteins for analysis

2. **In gel electrophoresis, where are the samples loaded?**
    - A. Into a transfer membrane
    - B. Into separate wells in the gel
    - C. Directly into the detection substrate
    - D. Into the blocking solution

    Answer: B. Into separate wells in the gel

3. **What is the purpose of transferring proteins to a solid support in Western blotting?**
    - A. To degrade the proteins
    - B. To visualize the proteins directly on the gel
    - C. For further analysis and detection of specific proteins
    - D. To prevent non-specific binding of antibodies

    Answer: C. For further analysis and detection of specific proteins

4. **Which type of transfer is recommended for large proteins (>100 kD)?**
    - A. Dry transfer
    - B. Wet transfer
    - C. Semi-dry transfer
    - D. Direct transfer

    Answer: B. Wet transfer

5. **Why is blocking an important step in the Western blotting procedure?**
    - A. It degrades non-specific proteins
    - B. It prevents non-specific binding of antibodies to the blotting membrane
    - C. It enhances the gel electrophoresis process
    - D. It labels the proteins for detection

    Answer: B. It prevents non-specific binding of antibodies to the blotting membrane

6. **What is the role of the primary antibody in Western blotting?**
    - A. To block non-specific binding
    - B. To transfer proteins to the membrane

- C. To detect and localize the protein of interest
- D. To label the proteins with HRP

Answer: C. To detect and localize the protein of interest

7. **What is the purpose of using a secondary antibody in Western blotting?**
   - A. To block non-specific binding
   - B. To transfer proteins to the membrane
   - C. To detect and localize the protein of interest
   - D. To bind to the primary antibody and facilitate detection

   Answer: D. To bind to the primary antibody and facilitate detection

8. **Which label is most commonly used on the secondary antibody in Western blotting?**
   - A. FITC
   - B. HRP
   - C. AP
   - D. Biotin

   Answer: B. HRP

9. **What happens when HRP (horseradish peroxidase) is exposed to a substrate solution in Western blotting?**
   - A. The proteins are transferred to the membrane
   - B. The proteins are blocked to prevent non-specific binding
   - C. The signal is detected, indicating the presence of the target protein
   - D. The gel electrophoresis process is enhanced

   Answer: C. The signal is detected, indicating the presence of the target protein

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1. **What is a DNA mutation?**
   - A. A temporary change in the nucleotide sequence of DNA
   - B. A permanent alteration of the nucleotide sequence of the genome
   - C. A reversible modification of genetic elements
   - D. A temporary shift in the structure of chromosomes

   Answer: B. A permanent alteration of the nucleotide sequence of the genome

2. **What can cause DNA mutations?**
   - A. Only errors in the replication process
   - B. Only damage to DNA that is not repaired
   - C. Damage to DNA, errors in replication, and insertion or deletion of DNA segments by mobile genetic elements

- D. Only insertion or deletion of DNA segments

Answer: C. Damage to DNA, errors in replication, and insertion or deletion of DNA segments by mobile genetic elements

3. **Do mutations always produce observable changes in an organism's phenotype?**
   - A. Yes, always
   - B. No, never
   - C. Sometimes, depending on the mutation
   - D. Only in viruses

   Answer: C. Sometimes, depending on the mutation

4. **In which biological processes do mutations play a role?**
   - A. Only in abnormal processes
   - B. Only in evolution
   - C. Both normal and abnormal processes, including evolution, cancer, and the development of the immune system
   - D. Only in the development of the immune system

   Answer: C. Both normal and abnormal processes, including evolution, cancer, and the development of the immune system

5. **What are the potential effects of mutations in genes?**
   - A. They always prevent the gene from functioning properly
   - B. They can have no effect, alter the product of a gene, or prevent the gene from functioning properly or completely
   - C. They always alter the product of a gene
   - D. They never affect non-genic regions

   Answer: B. They can have no effect, alter the product of a gene, or prevent the gene from functioning properly or completely

6. **Which of the following can damage DNA?**
   - A. Only chemical agents
   - B. Only physical agents
   - C. Both chemical and physical agents, which can be endogenous or exogenous
   - D. Only exogenous agents

   Answer: C. Both chemical and physical agents, which can be endogenous or exogenous

7. **Why is DNA stability important?**
   - A. To ensure accurate genetic information transfer between generations and proper gene function
   - B. To drive evolution
   - C. To prevent all types of genetic variation

- D. To increase mutation rates

Answer: A. To ensure accurate genetic information transfer between generations and proper gene function

8. **Why is genetic variation necessary?**
   - A. To ensure DNA stability
   - B. To prevent the formation of new species
   - C. To drive evolution and contribute to biodiversity
   - D. To eliminate endogenous agents

   Answer: C. To drive evolution and contribute to biodiversity

9. **What does the balance between DNA damage (mutation) and repair contribute to?**
   - A. Only the stability of DNA
   - B. Only the prevention of diseases
   - C. Both the continuity of genetic information and the driving force of evolution and biodiversity
   - D. Only the repair mechanisms

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1. **What is the simplest type of DNA mutation?**
   - A. Single base change
   - B. Extensive insertion
   - C. Gross rearrangement of chromosome structure
   - D. Large deletion

   Answer: A. Single base change

2. **What are the two kinds of single base change mutations?**
   - A. Deletions and Insertions
   - B. Transversions and Translocations
   - C. Transitions and Transversions
   - D. Transposons and Deletions

   Answer: C. Transitions and Transversions

3. **What is a transition mutation?**
   - A. A substitution of a pyrimidine for a purine
   - B. A substitution of a purine for a pyrimidine
   - C. A pyrimidine-to-pyrimidine or purine-to-purine substitution
   - D. An insertion of a transposon

   Answer: C. A pyrimidine-to-pyrimidine or purine-to-purine substitution

4. **Which of the following is an example of a transition mutation?**
    - A. Thymine (T) to Guanine (G)
    - B. Adenine (A) to Cytosine (C)
    - C. Thymine (T) to Cytosine (C)
    - D. Guanine (G) to Thymine (T)

    Answer: C. Thymine (T) to Cytosine (C)

5. **What is a transversion mutation?**
    - A. A substitution of a pyrimidine for a pyrimidine
    - B. A substitution of a purine for a purine
    - C. A pyrimidine-to-purine or purine-to-pyrimidine substitution
    - D. A small insertion or deletion

    Answer: C. A pyrimidine-to-purine or purine-to-pyrimidine substitution

6. **Which of the following is an example of a transversion mutation?**
    - A. Thymine (T) to Cytosine (C)
    - B. Adenine (A) to Guanine (G)
    - C. Thymine (T) to Guanine (G)
    - D. Cytosine (C) to Thymine (T)

    Answer: C. Thymine (T) to Guanine (G)

7. **What are point mutations?**
    - A. Mutations that cause large deletions
    - B. Mutations that alter a single nucleotide
    - C. Mutations that involve transposon insertions
    - D. Mutations that rearrange chromosome structure

    Answer: B. Mutations that alter a single nucleotide

8. **What kind of mutations cause extensive insertions, deletions, or gross rearrangements of chromosome structure?**
    - A. Point mutations
    - B. Transition mutations
    - C. Transversion mutations
    - D. Complex mutations

    Answer: D. Complex mutations

9. **What is a transposon and how does it affect DNA?**
    - A. A single base change that causes point mutations
    - B. A chemical agent that damages DNA
    - C. A genetic element that inserts many thousands of nucleotides of foreign DNA into coding or regulatory sequences

- D. A process of purine-to-pyrimidine substitution

Answer: C. A genetic element that inserts many thousands of nucleotides of foreign DNA into coding or regulatory sequences

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1. **What is biotechnology primarily concerned with?**
   - A. The use of chemicals to transform materials
   - B. The use of microbes, animal/plant cells, and their products to synthesize, break down, or transform materials
   - C. The use of traditional agricultural methods
   - D. The development of mechanical processes

   Answer: B. The use of microbes, animal/plant cells, and their products to synthesize, break down, or transform materials

2. **What technology is primarily included in biotechnology for improving processes?**
   - A. Mechanical engineering
   - B. Civil engineering
   - C. Recombinant DNA technology and genetic engineering
   - D. Traditional farming techniques

   Answer: C. Recombinant DNA technology and genetic engineering

3. **What is traditional biotechnology?**
   - A. Techniques used for modern genetic modification
   - B. Conventional techniques used for centuries to produce products like beer, wine, and cheese
   - C. Development of industrial processes
   - D. Use of cell fusion techniques

   Answer: B. Conventional techniques used for centuries to produce products like beer, wine, and cheese

4. **What does modern biotechnology encompass?**
   - A. Only traditional biotechnological processes
   - B. All methods of genetic modification by recombinant DNA and cell fusion techniques along with modern developments of traditional biotechnological processes
   - C. Only enzyme production
   - D. Mechanical processes in agriculture

   Answer: B. All methods of genetic modification by recombinant DNA and cell fusion techniques along with modern developments of traditional biotechnological processes

5. **What is white biotechnology focused on?**
    - A. Development of agricultural methods
    - B. Production of pharmaceuticals
    - C. Development of processes and microorganisms for industrial processes, such as enzyme production
    - D. Techniques for traditional food production

    Answer: C. Development of processes and microorganisms for industrial processes, such as enzyme production

6. **Which of the following is an example of traditional biotechnology?**
    - A. Genetic modification of crops
    - B. Production of beer and cheese
    - C. Recombinant DNA technology
    - D. Development of industrial enzymes

    Answer: B. Production of beer and cheese

7. **Which technique is used in modern biotechnology to modify genetic material?**
    - A. Traditional fermentation
    - B. Mechanical engineering
    - C. Recombinant DNA technology and cell fusion
    - D. Simple breeding techniques

    Answer: C. Recombinant DNA technology and cell fusion

8. **What is a primary application of white biotechnology?**
    - A. Creating genetically modified organisms
    - B. Traditional brewing of beer
    - C. Producing industrial enzymes
    - D. Developing pharmaceuticals

    Answer: C. Producing industrial enzymes

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1. **What is biotechnology primarily used for?**
    - A. Only for producing goods
    - B. Only for providing services
    - C. Producing goods, providing services, and for research and development
    - D. Only for agricultural purposes

    Answer: C. Producing goods, providing services, and for research and development

2. **How can biotechnology contribute to agriculture?**

- A. By replacing all traditional farming methods
- B. By optimizing industrial processes and helping agriculture adapt to climate change
- C. By eliminating the need for genetic modification
- D. By only focusing on increasing food production

Answer: B. By optimizing industrial processes and helping agriculture adapt to climate change

3. **What role does biotechnology play in dentistry?**
   - A. It eliminates the need for dental care
   - B. It replaces artificial materials with biological materials based on stem cells
   - C. It only focuses on cosmetic dental treatments
   - D. It uses only traditional dental techniques

   Answer: B. It replaces artificial materials with biological materials based on stem cells

4. **Which type of stem cells have been identified for potential use in dental applications?**
   - A. Only bone marrow mesenchymal stem cells (BMSC)
   - B. Only mesenchymal stem cells derived from adipose tissue (AD-MSCs)
   - C. Multiple types, including DPSCs, PDLSCs, SCAPs, SHED, PAFSCs, AD-MSCs, and BMSC
   - D. Only dental pulp stem cells (DPSCs)

   Answer: C. Multiple types, including DPSCs, PDLSCs, SCAPs, SHED, PAFSCs, AD-MSCs, and BMSC

5. **What significant contribution did biotechnology make during the COVID-19 pandemic?**
   - A. It developed new traditional farming techniques
   - B. It was used to develop the COVID-19 vaccine in record time
   - C. It eliminated the need for medical research
   - D. It focused on producing artificial food

   Answer: B. It was used to develop the COVID-19 vaccine in record time

6. **What are some of the ambitious goals laid out by the United Nations' Sustainable Development Goals (SDG) that biotechnology can help achieve?**
   - A. Only eradicating poverty
   - B. Only developing new pharmaceutical products
   - C. Combating hunger, achieving food security, promoting sustainable agriculture, and combating climate change
   - D. Only focusing on environmental protection

Answer: C. Combating hunger, achieving food security, promoting sustainable agriculture, and combating climate change

7. **What is the broad definition of biotechnology?**
    - A. Any technological application that uses chemical systems to create products
    - B. Any technological application that uses biological systems, living organisms, or parts/derivatives thereof to create or modify products and processes for specific uses
    - C. Only the use of genetic engineering techniques
    - D. Only traditional methods of making beer, wine, cheese, and bread

    Answer: B. Any technological application that uses biological systems, living organisms, or parts/derivatives thereof to create or modify products and processes for specific uses

8. **What are some applications of biotechnology in recent decades?**
    - A. Only traditional food production
    - B. Only pharmaceutical development
    - C. Pharmaceutical development, food production, and treatment of contaminants
    - D. Only agricultural processes

    Answer: C. Pharmaceutical development, food production, and treatment of contaminants

9. **What are some research goals of the Department of Microbial and Plant Biotechnology at the CIB Margarita Salas?**
    - A. Only focusing on plant resilience
    - B. Understanding how plants, arthropods, and microorganisms interact and respond to their environment to develop biotechnological applications for various sectors
    - C. Eliminating the need for biotechnology in agriculture
    - D. Only studying microbiological systems

    Answer: B. Understanding how plants, arthropods, and microorganisms interact and respond to their environment to develop biotechnological applications for various sectors

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1. **What is immunity?**
    - A. A state of vulnerability to infectious disease
    - B. A state of protection from infectious disease
    - C. A state of increased susceptibility to pathogens
    - D. A state of constant illness

    Answer: B. A state of protection from infectious disease

2. **What are the two components of immunity?**
    - A. Active and Passive Immunity
    - B. Innate and Adaptive Immunity
    - C. Primary and Secondary Immunity
    - D. Specific and Non-specific Immunity

    Answer: B. Innate and Adaptive Immunity

3. **What is the characteristic of innate immunity?**
    - A. It provides a specific response to pathogens
    - B. It provides the first line of defense against infection
    - C. It comes into play only after an antigenic challenge
    - D. It has memory for pathogens

    Answer: B. It provides the first line of defense against infection

4. **When does adaptive immunity come into play?**
    - A. Immediately upon infection
    - B. Only after an antigenic challenge to the organism
    - C. During the first line of defense
    - D. Before innate immunity

    Answer: B. Only after an antigenic challenge to the organism

5. **What is the primary function of the immune system?**
    - A. To promote the growth of pathogenic microorganisms
    - B. To protect animals from invading pathogenic microorganisms and cancer
    - C. To increase susceptibility to infections
    - D. To decrease the body's defense mechanisms

    Answer: B. To protect animals from invading pathogenic microorganisms and cancer

6. **What are the two related activities of the immune response?**
    - A. Recognition and Elimination
    - B. Recognition and Response
    - C. Response and Neutralization
    - D. Elimination and Prevention

    Answer: B. Recognition and Response

7. **What is the effector response in the immune system?**
    - A. The initial recognition of a pathogen
    - B. The recruitment of cells and molecules to eliminate or neutralize the organism
    - C. The memory response to a pathogen
    - D. The ability to discriminate between foreign molecules and the body's own cells

Answer: B. The recruitment of cells and molecules to eliminate or neutralize the organism

8. **What characterizes the memory response of the immune system?**
    - A. A slower and less effective immune reaction
    - B. A more rapid and heightened immune reaction upon later exposure to the same foreign organism
    - C. The initial exposure to a pathogen
    - D. The inability to recognize the same pathogen

    Answer: B. A more rapid and heightened immune reaction upon later exposure to the same foreign organism

9. **What is the source of the English word "immunity"?**
    - A. The Greek term "immunis"
    - B. The Latin term "immunis"
    - C. The French term "immunis"
    - D. The Spanish term "immunis"

    Answer: B. The Latin term "immunis"

10. **What observation led to the growth of the discipline of immunology?**
    - A. Individuals who had never been exposed to infectious diseases were protected
    - B. Individuals who had recovered from certain infectious diseases were thereafter protected from the disease
    - C. Animals who had strong innate immunity were more susceptible to diseases
    - D. Pathogens that were not recognized by the immune system were always neutralized

    Answer: B. Individuals who had recovered from certain infectious diseases were thereafter protected from the disease

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1. **What is the primary role of innate immunity?**
    - A. To provide a specific response to pathogens
    - B. To act as the first line of defense against infection
    - C. To create immunologic memory
    - D. To eliminate specific foreign microorganisms

    Answer: B. To act as the first line of defense against infection

2. **Which of the following is a characteristic of innate immunity?**
    - A. It is specific to a particular pathogen
    - B. It includes phagocytic cells, barriers like skin, and antimicrobial compounds

- C. It requires an antigenic challenge to activate
- D. It has immunologic memory

Answer: B. It includes phagocytic cells, barriers like skin, and antimicrobial compounds

3. **How does innate immunity differ from adaptive immunity in terms of specificity?**
   - A. Innate immunity is highly specific to each pathogen
   - B. Innate immunity is not specific to particular pathogens, while adaptive immunity is specific
   - C. Both are non-specific
   - D. Both are specific to each pathogen

   Answer: B. Innate immunity is not specific to particular pathogens, while adaptive immunity is specific

4. **What are the four characteristic attributes of adaptive immunity?**
   - A. First line defense, diversity, antigenic specificity, and immunologic memory
   - B. Self/nonself-recognition, antigenic specificity, diversity, and immunologic memory
   - C. Phagocytic activity, antigenic specificity, diversity, and barriers
   - D. Immediate response, diversity, antigenic specificity, and barriers

   Answer: B. Self/nonself-recognition, antigenic specificity, diversity, and immunologic memory

5. **Which cells are involved in innate immunity?**
   - A. T cells and B cells
   - B. Macrophages and neutrophils
   - C. Only neutrophils
   - D. Only macrophages

   Answer: B. Macrophages and neutrophils

6. **What is the primary advantage of adaptive immunity over innate immunity?**
   - A. It provides a uniform response in all members of a species
   - B. It can recognize and selectively eliminate specific foreign microorganisms and molecules
   - C. It acts faster than innate immunity
   - D. It does not require an antigenic challenge

   Answer: B. It can recognize and selectively eliminate specific foreign microorganisms and molecules

7. **Why does innate immunity act as the first line of defense?**
   - A. Because it is highly specific
   - B. Because it is present before the onset of infection

- C. Because it creates immunologic memory
- D. Because it requires time to marshal

Answer: B. Because it is present before the onset of infection

8. **What is an effector response in the context of the immune system?**
   - A. The initial recognition of a pathogen
   - B. The recruitment of cells and molecules to eliminate or neutralize the organism
   - C. The ability to distinguish between foreign molecules and the body's own cells
   - D. The immunologic memory response

   Answer: B. The recruitment of cells and molecules to eliminate or neutralize the organism

9. **Which attribute of adaptive immunity allows it to respond more effectively upon subsequent exposures to the same pathogen?**
   - A. Diversity
   - B. Immunologic memory
   - C. Antigenic specificity
   - D. Self/nonself-recognition

   Answer: B. Immunologic memory

10. **What does the antigenic specificity of the immune system allow it to do?**
    - A. Recognize a wide variety of pathogens non-specifically
    - B. Distinguish subtle differences among antigens
    - C. Provide a uniform response to all antigens
    - D. Eliminate the need for adaptive immunity

    Answer: B. Distinguish subtle differences among antigens

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1. **What is the primary function of the immune system?**
   - A. To help the body gain weight
   - B. To help the body fight infections and other diseases
   - C. To regulate body temperature
   - D. To improve digestion

   Answer: B. To help the body fight infections and other diseases

2. **What is an infection?**
   - A. The body's response to sunlight
   - B. The multiplication of germs such as bacteria or viruses in the body
   - C. The process of digestion

- D. The increase in body temperature

Answer: B. The multiplication of germs such as bacteria or viruses in the body

3. **Which part of the immune system helps prevent germs from getting into the body?**
   - A. White blood cells
   - B. Mucous membranes
   - C. Skin
   - D. Bone marrow

   Answer: C. Skin

4. **What are mucous membranes and their function in the immune system?**
   - A. The outer layer of skin; they provide protection against UV rays
   - B. Moist inner linings of some organs and body cavities that make mucus and other substances to trap and fight germs
   - C. Specialized cells that store fat
   - D. Hard tissues that protect internal organs

   Answer: B. Moist inner linings of some organs and body cavities that make mucus and other substances to trap and fight germs

5. **Which cells in the immune system are responsible for fighting germs?**
   - A. Red blood cells
   - B. Nerve cells
   - C. White blood cells
   - D. Muscle cells

   Answer: C. White blood cells

6. **What role do organs and tissues of the lymph system play in the immune system?**
   - A. They produce, store, and carry red blood cells
   - B. They produce, store, and carry white blood cells
   - C. They help in the digestion of food
   - D. They regulate the body's temperature

   Answer: B. They produce, store, and carry white blood cells

7. **What are antigens?**
   - A. Proteins that help in muscle contraction
   - B. Substances that the immune system sees as harmful or foreign
   - C. Hormones that regulate growth
   - D. Vitamins that support bone health

   Answer: B. Substances that the immune system sees as harmful or foreign

8. **What is an immune response?**
    - A. The body's response to cold temperatures
    - B. The process of digesting food
    - C. The immune system’s attack on recognized antigens
    - D. The production of red blood cells

    Answer: C. The immune system’s attack on recognized antigens

9. **What are antibodies and their function in the immune system?**
    - A. Proteins that help in digestion
    - B. Proteins that work to attack, weaken, and destroy antigens
    - C. Cells that transport oxygen
    - D. Hormones that regulate metabolism

    Answer: B. Proteins that work to attack, weaken, and destroy antigens

10. **What is immunity?**
    - A. The body's ability to gain muscle mass
    - B. Protection against a certain disease by recognizing and quickly responding to a previously encountered antigen
    - C. The process of breaking down food for energy
    - D. The regulation of blood sugar levels

    Answer: B. Protection against a certain disease by recognizing and quickly responding to a previously encountered antigen

11. **What is innate immunity?**

    - A. Immunity that develops after vaccination
    - B. Immunity that you are born with
    - C. Immunity that involves receiving antibodies from another source
    - D. Immunity that is specific to one type of pathogen

    Answer: B. Immunity that you are born with

12. **Which of the following is a characteristic of innate immunity?**

    - A. It develops after exposure to a foreign substance
    - B. It includes barriers like skin and mucous membranes
    - C. It involves the production of antibodies by the immune system
    - D. It provides long-lasting protection

    Answer: B. It includes barriers like skin and mucous membranes

13. **What is active immunity?**

- A. Immunity that you are born with
- B. Immunity that involves receiving antibodies from another source
- C. Immunity that develops after infection or vaccination
- D. Immunity that lasts only a few weeks or months

Answer: C. Immunity that develops after infection or vaccination

14. **Which type of immunity is usually long-lasting and can sometimes last a lifetime?**

- A. Innate immunity
- B. Passive immunity
- C. Active immunity
- D. Temporary immunity

Answer: C. Active immunity

15. **What is passive immunity?**

- A. Immunity that you develop after infection
- B. Immunity that involves receiving antibodies from another source
- C. Immunity that you are born with
- D. Immunity that lasts a lifetime

Answer: B. Immunity that involves receiving antibodies from another source

16. **How do newborn babies acquire passive immunity?**

- A. Through vaccination
- B. By developing their own antibodies
- C. Through antibodies received from their mothers
- D. Through infection with a pathogen

Answer: C. Through antibodies received from their mothers

17. **Which type of immunity provides immediate protection but is short-lived?**

- A. Innate immunity
- B. Active immunity
- C. Adaptive immunity
- D. Passive immunity

Answer: D. Passive immunity

18. **How can people acquire passive immunity other than from their mothers?**

- A. Through infection

- B. Through vaccination
- C. Through blood products that contain antibodies
- D. By producing their own antibodies

Answer: C. Through blood products that contain antibodies

19. **Which type of immunity includes cells and chemicals that can attack foreign substances?**

    - A. Passive immunity
    - B. Innate immunity
    - C. Active immunity
    - D. None of the above

    Answer: B. Innate immunity

20. **Which type of immunity is also called adaptive immunity?**

    - A. Innate immunity
    - B. Active immunity
    - C. Passive immunity
    - D. Natural immunity

    Answer: B. Active immunity

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1. **What can happen if the immune system overreacts to a non-threatening substance?**
    - A. Immunodeficiency diseases
    - B. Autoimmune diseases
    - C. Allergies and asthma
    - D. Cancer

    Answer: C. Allergies and asthma

2. **What characterizes an autoimmune disease?**
    - A. The immune system attacks pathogens
    - B. The immune system attacks healthy cells by mistake
    - C. The immune system fails to recognize foreign substances
    - D. The immune system produces too many white blood cells

    Answer: B. The immune system attacks healthy cells by mistake

3. **What is an immunodeficiency disease?**
    - A. A condition where the immune system overreacts to harmless substances
    - B. A genetic disorder where the immune system is weakened, leading to frequent and severe infections
    - C. An autoimmune condition where the immune system attacks itself
    - D. A disease that causes inflammation of the immune system

    Answer: B. A genetic disorder where the immune system is weakened, leading to frequent and severe infections

4. **What is a common effect of immunodeficiency diseases?**
    - A. Increased ability to fight infections
    - B. Frequent and severe infections that are harder to treat
    - C. Overproduction of immune cells
    - D. Rapid healing of wounds

    Answer: B. Frequent and severe infections that are harder to treat

5. **How does HIV affect the immune system?**
    - A. It increases the production of white blood cells
    - B. It destroys white blood cells
    - C. It enhances immune system function
    - D. It attacks red blood cells

    Answer: B. It destroys white blood cells

6. **What can untreated HIV lead to?**
    - A. Immunodeficiency disease
    - B. Allergies
    - C. AIDS (Acquired Immunodeficiency Syndrome)
    - D. Autoimmune diseases

    Answer: C. AIDS (Acquired Immunodeficiency Syndrome)

7. **What is the effect of AIDS on the immune system?**
    - A. It improves the immune system's ability to fight infections
    - B. It leads to a well-functioning immune system
    - C. It causes severe damage to the immune system, leading to increased susceptibility to severe illnesses
    - D. It prevents the immune system from attacking pathogens

    Answer: C. It causes severe damage to the immune system, leading to increased susceptibility to severe illnesses

8. **Which condition is characterized by the immune system attacking the body's own cells?**

- A. Allergies
- B. Autoimmune disease
- C. Immunodeficiency disease
- D. HIV/AIDS

Answer: B. Autoimmune disease

9. **What kind of diseases are often genetic disorders affecting the immune system?**
   - A. Allergies
   - B. Autoimmune diseases
   - C. Immunodeficiency diseases
   - D. Infectious diseases

   Answer: C. Immunodeficiency diseases

10. **What can result from an overactive immune response to non-threatening substances?**
    - A. Increased resistance to infections
    - B. Autoimmune diseases
    - C. Allergies and asthma
    - D. Normal immune function

    Answer: C. Allergies and asthma

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1. **What is the primary function of the lymphatic system?**
   - A. To produce hormones
   - B. To collect and filter lymph fluid from tissues
   - C. To digest food
   - D. To transport oxygen to cells

   Answer: B. To collect and filter lymph fluid from tissues

2. **What happens to lymph fluid in the lymphatic system?**
   - A. It is converted into blood
   - B. It is filtered to remove waste and germs
   - C. It is stored as fat
   - D. It is used to produce hormones

   Answer: B. It is filtered to remove waste and germs

3. **What are lymph nodes, and what do they do?**
   - A. They produce hormones that regulate growth
   - B. They filter lymph fluid and can swell during infection

- C. They transport oxygen to the bloodstream
- D. They digest food

Answer: B. They filter lymph fluid and can swell during infection

4. **Which types of white blood cells are produced in the lymphatic system?**
   - A. Red blood cells and platelets
   - B. T lymphocytes (T cells) and B lymphocytes (B cells)
   - C. Neutrophils and eosinophils
   - D. Monocytes and basophils

   Answer: B. T lymphocytes (T cells) and B lymphocytes (B cells)

5. **Where are T lymphocytes and B lymphocytes most likely to encounter foreign microorganisms?**
   - A. In the bloodstream
   - B. In the lymphatic system
   - C. In the digestive system
   - D. In the respiratory system

   Answer: B. In the lymphatic system

6. **What does swelling of lymph nodes usually indicate?**
   - A. A deficiency in white blood cells
   - B. A healthy immune response
   - C. An infection or inflammation
   - D. A problem with the circulatory system

   Answer: C. An infection or inflammation

7. **What is the role of lymphocytes in the immune system?**
   - A. To carry nutrients to tissues
   - B. To help rid the body of disease-causing microorganisms
   - C. To transport oxygen to the cells
   - D. To break down food in the stomach

   Answer: B. To help rid the body of disease-causing microorganisms

8. **Which part of the lymphatic system helps produce, differentiate, and proliferate lymphocytes?**
   - A. The skin
   - B. The lymph nodes
   - C. The organs and tissues of the lymphatic system
   - D. The liver

   Answer: C. The organs and tissues of the lymphatic system

9. **How does the lymphatic system contribute to the immune response?**
    - A. By digesting pathogens
    - B. By producing and storing red blood cells
    - C. By filtering lymph and producing lymphocytes
    - D. By transporting hormones

    Answer: C. By filtering lymph and producing lymphocytes

10. **What can cause lymph nodes to become noticeably swollen?**
    - A. A cold drink
    - B. Excessive exercise
    - C. An infection
    - D. Lack of sleep

    Answer: C. An infection